AI Video Director

Skills, method and sources

Digikon Solutions | 20 October 2026 | Design proposal. Implementation has not started and no integration described here has been tested yet.

1. Purpose

The AI Video Director takes a client brief and assets through creative planning, validated editing, rendering, quality checks and targeted revision. It uses **one primary Director agent** (Claude) that applies **seven specialist skills**, retrieves supporting knowledge through **RAG**, and chooses from a **library of validated recipes** instead of inventing editing operations. Application code validates every plan before Remotion and FFmpeg execute it.

2. The seven skills and where each comes from

Skill	What it does	How it is applied	Knowledge sources
video-director	Coordinates the others: reads the brief, routes work, resolves conflicts, runs revisions	Loads only the skills needed at each step; stops after two revision loops and escalates to a human	Original; follows the Agent Skills format
commercial-strategy	Message, hook, structure, call to action, claims, platform fit	Produces the treatment and script; every claim is linked to an approved brand fact	Original; ad-structure references in section 6
cinematography	Shot choice and framing from the supplied footage	Selects usable segments and reframes them; flags shots the footage cannot support	Original; film-craft references in section 6
editing-director	Pacing, cuts, transitions, timeline	Turns the storyboard into typed edit operations taken from the recipe library	Original; editing references in section 6; video-shotcraft sequences
motion-graphics	On-screen text, graphics, animation, caption styling, safe areas	Picks animation recipes and fills their parameters within allowed ranges	video-shotcraft shot cards; Remotion agent skills
sound-design	Music and sound-effect choice, cue timing, ducking, loudness	Cue times are tied to scene IDs; only licensed audio is used	Original; EBU R 128 loudness reference
video-qa	Inspects the rendered video and reports problems with evidence	Deterministic checks in code plus Claude vision on sampled frames; each finding becomes a revision ticket	Original; FFmpeg analysis filters (section 6)

Skills are short instruction files (SKILL.md) with detail in a references folder and test cases in an evals folder. Where no vetted external skill exists, we write an original one; external material is used as a reference, never copied in unreviewed.

3. How skills, RAG and the Director work together

Skills define how a task is done. RAG supplies the facts, recipes and examples needed for that step. The full knowledge base is never placed in a prompt.

Step	Skills used	RAG retrieves	Result
Intake	video-director	Verified platform specs	Validated brief
Asset analysis	cinematography	None	Asset notes, uncertainty flagged
Treatment and script	commercial-strategy	Principles, playbooks, brand facts	Treatment, claims register
Storyboard	commercial-strategy, cinematography	Principles, platform specs	Scenes with shot intent
Edit plan	editing-director, motion-graphics, sound-design	Recipes filtered by type, compatibility, duration	Typed operations with recipe IDs
Validate, then approve	Application code, then human	None	Plan passes checks and is approved

Step	Skills used	RAG retrieves	Result
Render	Remotion and FFmpeg	None	Scene clips and preview
QA	video-qa	QA criteria, brand facts	Findings with evidence
Revise	video-director plus the owning skill	Recipes, findings	Only the affected scenes change

- **Collections:** directing principles, ad playbooks, platform specs (verified only), shot recipes (reviewed only), brand and product facts (per project), approved examples (added after review).
- **Retrieval:** metadata filter, then dense plus sparse search fused with RRF, then rerank (Qdrant hybrid search). Chunks carry added context (Contextual Retrieval). Results are benchmarked against a keyword baseline on a curated question set before the approach is fixed.
- **Conflict priority:** legal text and approved claims, product fidelity, platform specs, message clarity, pacing, style.

4. Recipe library

Each recipe is a validated record: ID and version, type (shot, camera move, transition, animation, sound cue), purpose, typed parameters with ranges, constraints (duration limits, maximum uses), compatibility (renderer, aspect ratio, media type), validation rules, and a link to a reviewed Remotion or FFmpeg template. The Director adapts recipes to the brief; the validator rejects any unknown recipe or out-of-range value. The starter set is 15 to 20 original recipes. Cards from video-shotcraft (Apache-2.0) are converted and imported one at a time after review.

5. Quality checks, revision and learning

- **Technical checks (code):** duration, resolution, frame rate, audio stream, black and frozen frames, silence, loudness, caption overflow.
- **Visual checks (Claude on sampled frames):** product and brand fidelity, pacing, continuity, transitions. No finding is accepted without evidence.
- **Revision:** a finding names scene IDs; only those scenes are revised and re-rendered, then checked again with their neighbours and the global checks.
- **Learning:** user feedback becomes a candidate lesson. A reviewer approves it and a regression suite compares old and new versions before any skill or recipe changes. Every export records the model, skill, recipe and tool versions.

6. Research flow and sources

Step	What we did
1. Baseline	Started from Digikon's internal architecture blueprint (Claude director, RAG, skills, LangGraph.js, deterministic media tools).
2. Existing skills	Searched for ready-made director and editing skills. None was suitable and vetted, so the director skills are written originally.
3. Craft research	Reviewed editing, cinematography, advertising-structure, motion-design and sound sources to define each skill's rules.
4. Recipe library study	Read the video-shotcraft library, a full recipe card and its repository to design our recipe schema.
5. Technical foundations	Reviewed the skill format, retrieval, orchestration, rendering and analysis tooling documentation.
6. Grading	Official and primary sources define firm rules. Industry and vendor articles are treated as hypotheses to be tested. Platform numbers are confirmed in each platform's official documentation before use.
7. Next	Write evaluation cases first, then skill files, retrieval and the pipeline.

Skills and agent design

Source (click)	Used for
Agent Skills overview (Claude Docs)	Skill file format and progressive loading
Skill authoring best practices (Claude Docs)	Concise skills, testing with real usage
Equipping agents with Agent Skills (Anthropic Engineering)	Metadata, body and reference loading levels

Shot recipes and rendering

Source (click)	Used for
video-shotcraft shot library	Shot-recipe categories and card structure
video-shotcraft repository	Production method and Remotion implementations
Example recipe card	Recipe fields: intent, parameters, pitfalls
Remotion agent skills	Best practices for Remotion markup, rendering and captions

Editing craft

Source (click)	Used for
Adobe: J-cut and L-cut	Split edits
Adobe Help: J and L cuts in Premiere Pro	How split edits are built in an editor
Cutback: jump cut, J-cut, L-cut	Cut-type vocabulary
No Film School: Murch's six cutting rules	Priority order for cut decisions
Murch's Rule of Six (thesis)	Academic write-up of the same criteria

Advertising strategy, cinematography and motion design

Source (click)	Used for
Benly: video ad structure frameworks	Hook and structure patterns
Hooked: script for an advertisement	Script sequence and length guidance
Oakgen: first-frame hooks	Opening-frame principles
CSUS: Elements of cinematography	Framing and continuity basics
Videomaker: the 180-degree rule	Continuity rule
GitLab Design: brand motion	Easing, hierarchy, purposeful motion

Retrieval, orchestration and quality tooling

Source (click)	Used for
Anthropic: Introducing Contextual Retrieval	Chunk context and reranking approach
Qdrant: hybrid search	Dense plus sparse search with RRF fusion
Qdrant: hybrid and multi-stage queries	Multi-stage retrieval and rerank
LangGraph.js persistence	Saved state and resuming interrupted runs
LangGraph.js human-in-the-loop	Approval pauses
FFmpeg filters documentation	Black, freeze, silence and loudness analysis
EBU R 128 loudness recommendation	Broadcast loudness reference (platform targets confirmed separately)

Note on confidence. Figures from articles, including retrieval-improvement percentages and social-platform safe-zone margins, are used as starting hypotheses and are confirmed against official documentation and our own benchmarks before they become rules.